

Otto Group Academy LMS

Features & Capabilities Reference

Product: Otto Group Academy Learning Management System
Audience: VECTRA International / Otto Group stakeholders, administrators, and technical teams
Version: Current MVP (as of August 2026)
Stack: Next.js · React · TypeScript · MongoDB · SCORM 1.2

1. Executive summary

Otto Group Academy LMS is a full-stack learning platform built for VECTRA International and Otto Group. It delivers SCORM 1.2 training to facility and business partner learners, tracks progress and completion, and gives administrators tools to manage rosters, users, courses, assignments, and compliance reporting.

The system supports self-registration and admin invitation flows, both gated by an approved participant roster. Content is authored externally (e.g. Mindsmith) and uploaded as SCORM 1.2 ZIP packages for reliable progress, score, and completion tracking.

2. User roles & permissions

Role	Access level
Learner	Dashboard, assigned courses, SCORM player, profile (name), password management
Coordinator	Admin console: learners, courses, assignments, reports. Cannot manage participant roster, create staff, or remove users
Admin	Full system control including participant roster, staff accounts, user removal, and all coordinator capabilities

Stakeholder groups (learners)

- **Facility** — organization linked to a facility training site
- **Business Partner** — company-level participation on the VECTRA roster

Registration and invitations require a valid **Company ID** and **stakeholder group** that match an entry on the approved participant roster.

3. Authentication & account management

Feature	Description
Email/password login	Secure sign-in with HTTP-only session cookies (12-hour sessions)
Self-registration	Learners register with corporate email, name, Company ID, stakeholder group, and organizational name
Roster validation	Registration succeeds only when Company ID, stakeholder group, and organizational name match the approved roster

Invitation flow	Admins/coordinators create users; system sends activation email with secure link
Account activation	Invited users set password via activation link (7-day token expiry)
Forgot / reset password	Email-based password recovery
User statuses	Invited, Active, Inactive
Resend invitation	Resend activation email to invited users
Bootstrap admins	System administrator accounts auto-provisioned from environment configuration
Logout	Secure session termination

Email delivery

Supports multiple providers (environment-driven):

- Resend API
- SendGrid API
- SMTP (Nodemailer)

Used for invitations, activation, and password reset. Activation links are never displayed in the admin UI.

4. Participant roster (approved organizations)

The participant roster is the source of truth for which organizations and Company IDs may register or be invited.

Feature	Admin	Coordinator
View roster	Yes	Yes (read-only)
Add organization	Yes	No
Edit organization	Yes	No
Remove organization	Yes	No
Import CSV / XLSX	Yes	No
Bulk remove selected	Yes	No

Roster fields

Field	Description
Stakeholder group	Facility or Business Partner
Company ID	Unique identifier used at registration
Organization name	Official name on the roster
Belongs to BP	Business partner affiliation (if applicable)
Country	Organization country

Topic	Training topic (e.g. Freely Chosen Employment)
Nominated provider	Training provider (VECTRA normalized to uppercase)

Import formats

- **CSV** — comma-separated values with header row
- **XLSX** — Excel spreadsheet (first worksheet used)

Required import headers: Stakeholder , ID , Name , Belongs to BP , Country , Topic , Nominated Provider .

List features

- Search across company ID, name, country, and other fields
- Filter by stakeholder group and country
- Pagination (10 / 25 / 50 per page)
- Summary stats: total organizations, facilities, business partners

5. User management (learners & staff)

Feature	Admin	Coordinator
Create & invite learner	Yes	Yes
Create staff (Coordinator / Admin)	Yes	No
Import learners (CSV / XLSX)	Yes	Yes
Edit user profile	Yes	Learners only
Activate / deactivate	Yes	Learners only
Resend invite	Yes	Yes (learners)
Remove user (single)	Yes	No
Bulk remove selected	Yes	No

Learner import formats

- **CSV** and **XLSX** supported
- Required headers: First Name , Last Name , Corporate Email , Company ID , Stakeholder Group , Organizational Name
- Company ID and stakeholder group must match the participant roster
- Invalid rows are skipped with error summary (up to 20 errors returned)

User list features

- Search by name, email, company ID, entity, role, status
- Filter by status (Invited, Active, Inactive) and role (Learner, Coordinator, Admin)
- Pagination
- Bulk selection with checkbox (select all on current page; selection persists across pages)

Protections

- Cannot remove your own account
 - Bootstrap administrator accounts cannot be removed
 - Duplicate email addresses rejected
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6. Course management

Feature	Description
Upload SCORM 1.2 ZIP	Upload packaged course with title and description
Edit course	Update title and description
Replace SCORM package	Upload new ZIP to replace content on existing course
Remove course	Deletes course and all learner assignments for that course
Course types	SCORM 1.2 (primary); Mindsmith link type defined in schema

SCORM package handling

- ZIP extraction and manifest launch path detection
- Secure same-origin content serving via API routes
- Storage backends:
 - **Local filesystem** — development and simple deployments
 - **Cloudflare R2** — persistent storage for cloud hosts (e.g. Render) where local disk is ephemeral

Supported SCORM capabilities

Capability	Supported
ZIP upload & manifest launch	Yes
Lesson status / completion	Yes
Score tracking	Yes
Bookmark / suspend data / resume	Yes
Progress percentage	Yes
SCORM 2004 sequencing	No

Recommendation: Author in Mindsmith → export as **SCORM 1.2 ZIP** → upload in Admin → Courses.

7. Course assignment

Feature	Description
Single assign	Assign one course to one learner
Unassign	Remove a course assignment from a learner
Bulk assign	Assign to multiple learners by filter

Auto-assign on registration	When learner registers, course with matching topic title is assigned automatically
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Bulk assign filters

- All active and invited learners
- Company ID
- Country
- Stakeholder group (Facility / Business Partner)

8. Learner experience

Dashboard (/dashboard)

- Welcome overview with assigned, in-progress, and completed course counts
- Progress summary bar
- Quick link to course list

My courses (/dashboard/courses)

- List of assigned courses with status badges
- Progress indicators
- Actions: **Start course**, **Continue course**, **Review course**

SCORM player (/learn/[assignmentId])

- Full in-browser SCORM 1.2 runtime
- LMS API bridge (Initialize, GetValue, SetValue, Commit, Finish)
- Automatic progress save on commit and exit
- Resume from last bookmark / suspend data
- Mindsmith postMessage progress bridge support
- Assignment status updates: Not Started → In Progress → Completed

Profile (/dashboard/profile)

- Update display name (no re-registration required)
- View read-only account details (email, entity, company ID, stakeholder group, role)

Assignment statuses

Status	Meaning
Not Started	Course assigned but not yet launched
In Progress	Learner has started; progress saved
Completed	Course finished per SCORM completion rules

9. Reporting & analytics

Progress reports (/admin/reports)

Column	Description
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Learner name	Full name
Email	Corporate email
Entity	Organization
Country	Learner country
Course	Course title
Status	Not Started / In Progress / Completed
Progress	Percentage
Score	SCORM score (when available)
Last activity	Timestamp of last progress save
Completed at	Completion timestamp

Report features

- Filter by status, course, and country
- Search across learner name, email, entity, course
- Pagination (25 per page default)
- **Export CSV** — downloads filtered report matching current filters and search

Admin dashboard stats

- Approved organizations count
- Active and invited learners count
- Assignment completion rate (%)

10. Admin console screens

Path	Purpose
/admin	Administration overview and onboarding checklist
/admin/participants	Participant roster management
/admin/users	Learner and staff management
/admin/courses	SCORM upload, course edit, assignment
/admin/reports	Progress reports and CSV export

Public / auth screens

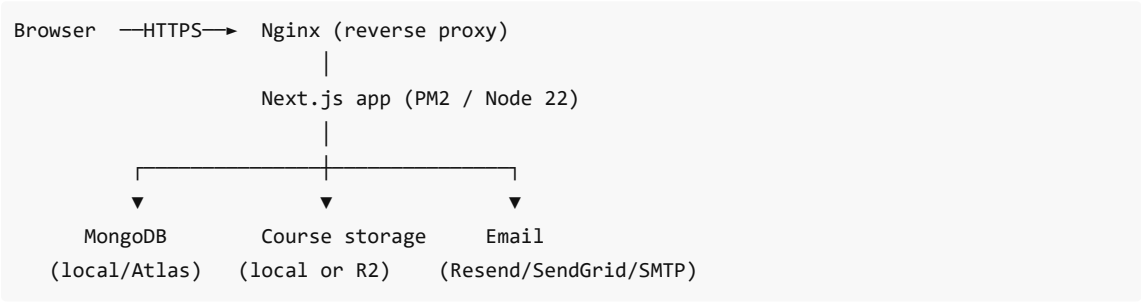
Path	Purpose
/login	Sign in
/register	Self-register (roster-gated)
/activate	Account activation

/forgot-password	Request password reset
/reset-password	Set new password

11. UI & branding

- Otto Group–styled application shell (header, footer, navigation)
- OTTO Group logo in header
- Partner logo strip: bonprix, OTTO, Crate&Barrel, Witt-Gruppe
- Responsive layout with mobile navigation menu
- Otto Group website search integration in header
- Branded login, registration, and activation pages

12. Technical architecture



Technology stack

Layer	Technology
Framework	Next.js 16 (App Router)
UI	React 19 + TypeScript
Database	MongoDB 7 driver
Authentication	jose (JWT) + HTTP-only cookies + bcrypt
Validation	Zod
SCORM	Custom SCORM 1.2 runtime + adm-zip
Spreadsheet import	csv-parse + xlsx (SheetJS)
Email	Nodemailer / Resend / SendGrid
Process manager	PM2
Node.js	22.x

Data collections

- **users** — accounts, roles, statuses, invite/reset tokens
- **participants** — approved organization roster

- **courses** — SCORM course metadata and launch paths
- **assignments** — learner–course links with progress and SCORM data

API endpoints (summary)

27 REST API routes covering:

- Authentication (login, logout, register, activate, forgot/reset password)
- User profile and admin user management (including import and bulk delete)
- Participant roster (CRUD, import, bulk delete)
- Course management and SCORM upload
- Assignments (single, bulk, unassign)
- SCORM content serving and launch
- Progress persistence
- Reports CSV export
- Facilities lookup (registration validation)
- Health check

13. Deployment options

Environment	Notes
Azure VM	Recommended pilot path: Ubuntu, Node 22, PM2, Nginx, Certbot, MongoDB
Render / cloud	Supported with Cloudflare R2 for SCORM storage persistence
Local development	<code>npm run dev</code> ; local filesystem for course storage

See also: `AZURE_DEPLOYMENT.md` , `docs/RENDER-SCORM-Storage.md` , `docs/Cloudflare-R2-Setup.md` .

14. Requirements coverage

Requirement	Status
Registered-user authentication	Covered
Learner CSV / XLSX import	Covered
90+ learners (no fixed cap)	Covered
Learner dashboard with status	Covered
SCORM 1.2 delivery + resume/completion	Covered
Mindsmith authoring path (via SCORM export)	Covered
Participant roster management	Covered
Administration (users, courses, assignments)	Covered
Reporting (UI + CSV export)	Covered
Bulk user/participant operations	Covered
Responsive UI	Covered

Azure VM deployment	Covered
Invitation email (SMTP / Resend / SendGrid)	Covered
Role-based access (Admin / Coordinator / Learner)	Covered

15. Known limitations

1. **SCORM 1.2 only** — not a certified SCORM engine; no SCORM 2004 sequencing
2. **Mindsmith published URLs** — suitable for preview only; not for operational progress reporting
3. **Same-VM MongoDB** — acceptable for pilot; Atlas + backups recommended for production scale
4. **Email deliverability** — production requires SPF/DKIM/DMARC configuration with Otto IT
5. **Package scanning** — no malware scan on uploaded SCORM ZIP files (recommended for untrusted uploaders)
6. **Audit trail** — no formal audit log UI yet (recommended for production compliance)

16. Recommended onboarding sequence (admin)

1. Import the VECTRA participant roster (Facilities / Business Partners + Company IDs) via CSV or XLSX
2. Upload the Freely Chosen Employment SCORM 1.2 course
3. Learners self-register with an approved Company ID, or import a learner CSV/XLSX file
4. Assign the course (auto-assigned on registration when course title matches topic)
5. Export progress reports as training completes

17. Related documentation

Document	Purpose
README.md	Project setup and quick start
docs/Otto-LMS-Presentation.md	Slide-style project presentation
docs/Learner-Guide.md	End-user learner instructions
docs/Mindsmith-SCORM-Progress-Tracking.md	SCORM progress integration details
docs/Mindsmith-SCORM-Delivery-Checklist.md	Content delivery checklist
docs/Cloudflare-R2-Setup.md	R2 storage configuration
docs/RENDER-SCORM-Storage.md	Render deployment storage notes
AZURE_DEPLOYMENT.md	Azure VM deployment runbook

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